

//Lab 2: write c program to Draw line using DDA.

```
#include <stdio.h>
#include <graphics.h>
#include <math.h>
int main()
{
    int x1, y1, x2, y2;
    float x, y, dx, dy;
    int steps, i;
    float x_inc, y_inc;
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "C:\\TURBOC3\\BGI");
    printf("Enter x1, y1: ");
    scanf("%d %d", &x1, &y1);
    printf("Enter x2, y2: ");
    scanf("%d %d", &x2, &y2);

    dx = x2 - x1;
    dy = y2 - y1;

    if (abs(dx) > abs(dy))
        steps = abs(dx);
    else
        steps = abs(dy);

    x = x1;
    y = y1;

    x_inc = dx / (float)steps;
    y_inc = dy / (float)steps;

    putpixel((int)(x+0.5), (int)(y+0.5), WHITE);
    for (i = 1; i <= steps; i++) {
        x = x + x_inc;
        y = y + y_inc;

        putpixel((int)(x+0.5), (int)(y+0.5), WHITE);
    }

    getch();
    closegraph();
    return 0;
}
```